

EE2300 Design Lab Report

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Introduction:

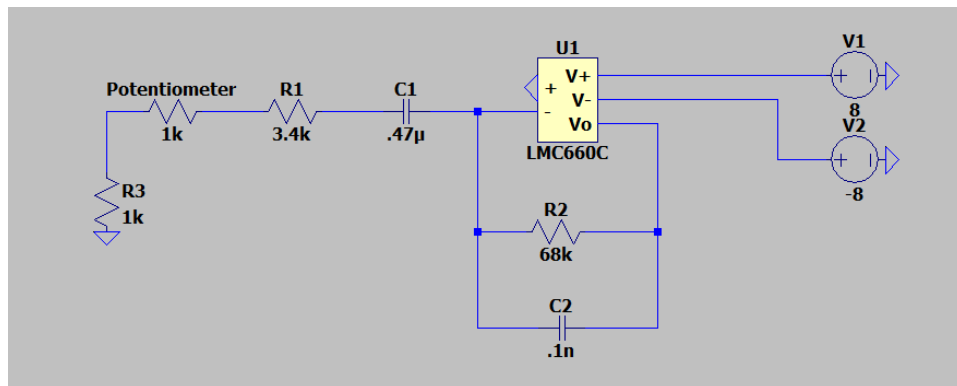
For the past few weeks, our lab group has been tasked with choosing a set of options of designs to choose from that we would have to build, test, and explain our results. Initially, our group chose to do the temperature controller project. We were able to figure out how to make the green, yellow, and red LEDs light up at the right time when the voltage was increased. However, we were unable to figure out how to make the LEDs turn off at different voltage values when the voltage was decreased. This frustrated us to now end, so we ended up changing our project to the two-channel headphone amplifier. We thought that this choice would be easier, but it still came with its own set of challenges.

Description of design and design process:

To start our process with this project, we took a look at the given parameters of the circuit. A few pieces of information were that the gain for each channel function had to be in the form of a bandpass response with lower and upper corner frequencies. Using our knowledge of circuits, we concluded that two bandpass filters were needed for this project to be successful. First, we had to find the value of the center frequency, f_0 . Next, we used the website, “Electronics Stack Exchange”, to figure out the relationship between the pass band gain and the resistor values when it came to a bandpass filter. Then, after deciding our resistor values using the ratio that we found, we used the website, “Electronics Tutorials”, to figure out the equations for each corner frequency that would lead us to finding the value of each capacitor.

After this initial process, it was time to build the circuit. We used the LMC660 operational amplifier for our bandpass filters. The power supplies were ± 12 volts. This value was just a guess but we figure that a value of ± 8 volts would also be viable. We figure that using the LM324 would also be a good choice for this design. We used two channels of the op-amp: one for the left ear, the other for the right. Now, let's go back to the project's given information for a second. One of the parameters was that a potentiometer should be used for each channel's volume control. We used the 102 potentiometers that have a maximum value of one kilohm. Also, it said to use a voltage divider to implement the potentiometers. With that being said, we made a complete guess for the value of the resistor that fulfills the voltage divider that implements the potentiometer. Let it be noted that different resistor values could work for this particular section, but we chose one kilohm. Also, we used a 5-volt DC power source to power our speaker for the circuit. This was inserted on the positive input strip of the circuit. Every calculation and equation that was mentioned in this section can be seen below in the rest of our lab report.

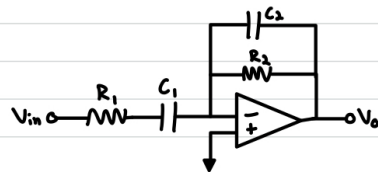
-Circuit Schematic (Same circuit for left and right channel)



-Design equations

$$f_{c,low} = 100 \text{ Hz} \quad f_{c,high} = 15 \text{ kHz} \quad G_{passband} = 20 \frac{\text{V}}{\text{V}}$$

Two bandpass filters



Passband Gain = gain at f_o

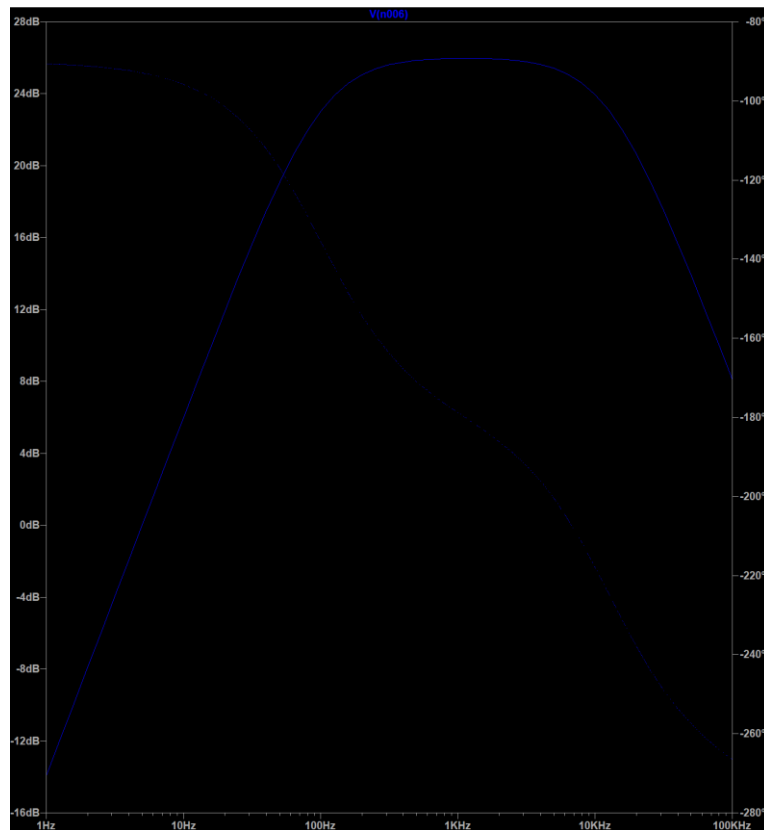
$$f_o = \sqrt{f_{low} f_{high}} = 1.225 \text{ kHz}$$

$$G_o = \frac{R_2}{2R_1} = 20 \rightarrow R_2 = 68\text{k} \quad R_1 = 3.4\text{k}$$

$$f_{c1} = \frac{1}{2\pi R_1 C_1} \rightarrow C_1 = \frac{1}{f_{c1} 2\pi R_1} = 0.47 \mu\text{F}$$

$$f_{c2} = \frac{1}{2\pi R_2 C_2} \rightarrow C_2 = \frac{1}{f_{c2} 2\pi R_2} = 0.15 \text{ nF} \approx 0.1 \text{ nF}$$

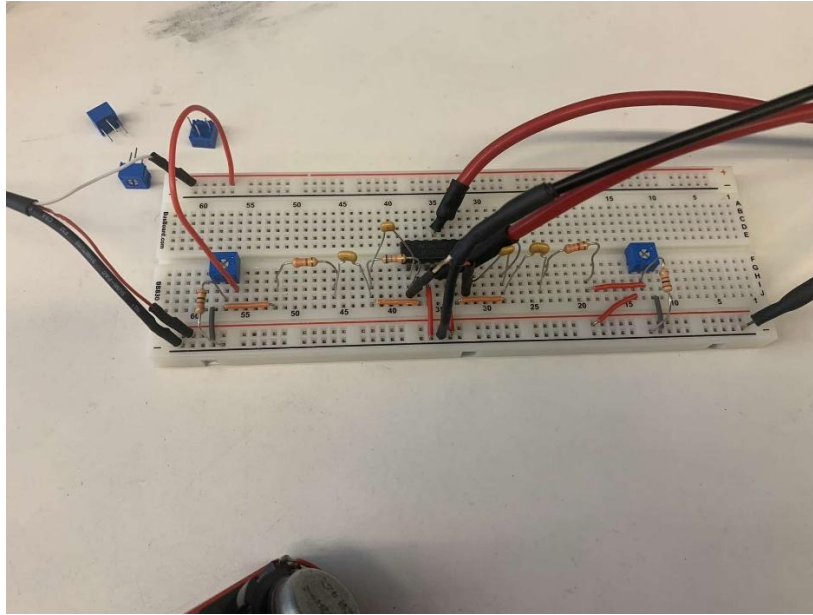
-Simulation results



-Required measurements described in project description

We had a few requirements for this project that had to be fulfilled. To start, we needed to use two potentiometers, one at each channel, made with voltage dividers that would serve as the volume control for each channel. Next, we needed a bandpass gain of 20 for each channel, along with a lower and upper corner frequency of 100 Hz and 15 kHz. Lastly, we were allowed to use up to two DC power supplied to power our circuit. These were separate from the power supplies used for the operational amplifier.

-Photograph of circuit



Conclusion:

In conclusion, we were able to design, build, and test this circuit successfully. We met all the requirements, as our corner frequencies were 99.6Hz and 15.6kHz. Also our gain was 20.26 V/V which was very close to the required 20V/V. We used a speaker from the 224 lab room and were able to hear our circuit work as well. Overall, we learned a lot during this design project and hope that this experience will help us in the future.

Bibliography:

“Active Band Pass Filter.” *Basic Electronics Tutorials*, 30 Apr. 2023, www.electronicstutorials.ws/filter/filter_7.html.

“Gain of Inverting Active Bandpass Filter.” *Electrical Engineering Stack Exchange*, 1 July 2019, electronics.stackexchange.com/questions/397981/gain-of-inverting-active-bandpass-filter?